

Deformable Object Simulation and Parameter Identification using Depth Observations

Antonio Škara, Bruno Maric, Frano Petric and Matko Orsag

Working draft — example sections; experimental results are not included.

Abstract—Predicting dough deformation during robotic manipulation requires a model that captures recoverable, rate-dependent, and persistent deformation without making calibration impractical. This paper presents TaichiDough, a framework for differentiable identification of effective simulation parameters from single-view depth observations and recorded rigid-tool trajectories. A table-aligned reconstruction initializes a three-dimensional moving least squares material point simulator with affine particle-grid transfer. The constitutive approximation combines fixed-corotated elasticity, an additive viscous term, and principal-stretch plastic projection. Scene geometry and marker-to-tool transforms are calibrated separately and held fixed during material identification. A partial-observation objective combines depth-change agreement, observed-region coverage, and directed distances to the predicted visible geometry, without treating unobserved space as known empty space. Checkpointed reverse-mode differentiation supports bounded optimization of Young’s modulus, Poisson’s ratio, viscosity, and two plastic stretch limits, with an extension to shared-material fitting across recordings. The central research question is whether time-resolved visual calibration improves prediction of unseen dough manipulation under fixed parameters. We distinguish this predictive objective from recovery of unique physical constants, since hidden volume, contact assumptions, and incomplete mechanical excitation can produce similar observations for different parameter combinations. The framework defines an evaluation protocol for testing that distinction; real-data predictive performance remains to be established.

I. INTRODUCTION

Dough manipulation combines large deformation, changing tool contact, incomplete observation, and a response that depends on preparation and loading history. These properties complicate the use of simulation for planning, control, and robot learning. A useful model must predict how an object will deform under a proposed action, but its initial geometry and mechanical parameters are rarely known with the accuracy assumed in a synthetic experiment. Deformable-object modeling therefore requires a joint consideration of numerical methods, sensing, and identification, rather than only a more detailed forward simulator [1].

All authors are with the LARICS Laboratory for Robotics and Intelligent Control Systems, University of Zagreb, Faculty of Electrical Engineering and Computing, Zagreb 10000, Croatia. (antonio.skara, bruno.maric, frano.petric, matko.orsag) at fer.hr.

For flour dough, this difficulty is particularly important. Elastic recovery, rate-dependent resistance, lasting deformation, and adhesion may occur within the same manipulation. Their relative importance varies with formulation, hydration, mixing, and previous handling [2]. A compact constitutive approximation can still be useful within a restricted operating range. However, minimizing an image discrepancy does not by itself establish that its parameters are intrinsic material properties. An apparent change in stiffness or viscosity can also compensate for an inaccurate initial thickness, incorrect tool pose, or excessive numerical dissipation.

Recent work has established several parts of this problem. RoboCraft and RoboCook demonstrate learned predictive models for elastoplastic manipulation and real dough tasks [3], [4]. Differentiable physics-based system identification (DPSI) estimates elastoplastic material parameters from robot-conditioned observations of plasticine [5]. EMPM includes visually calibrated material point simulation and bimanual bread-dough interaction [6]. Single-depth-camera calibration is also established for soft solids with known reference geometry [7]. Consequently, the combination of depth sensing, two tools, and material point simulation is not sufficient as a novelty claim. The relevant question concerns the information provided by a measurement protocol and the predictive value of the resulting model.

We investigate the following hypothesis: *a simplified elastoplastic model, calibrated from time-resolved observations of a single depth view and recorded tool trajectories, can improve prediction of previously unseen dough manipulation without further material-parameter adjustment.* Recorded tool motion supplies the prescribed mechanical input; the method is not a camera-only reconstruction of unknown actions. The observation sequence supplies intermediate deformation information, potentially distinguishing parameter combinations that produce similar endpoints. Whether it actually does so must be tested against endpoint-only fitting and under different loading and recovery conditions.

TaichiDough implements this identification problem using moving least squares material point simulation, a partial-visible-geometry objective, and checkpointed reverse-mode differentiation. Three methodological elements organize the study. First, scene alignment, tool registration, and hidden-volume reconstruction are made explicit and separated from material optimization. Second, a bounded five-parameter model is fitted using a loss that

accounts for incomplete depth support while penalizing missing predictions in observed regions. Third, shared-material fitting and frozen-parameter evaluation distinguish agreement on calibration recordings from prediction on independent episodes.

The intended contribution is an experimentally testable approach to measurement-conditioned dough modeling, not a new continuum discretization or proof of unique rheological identification. The present draft specifies the implemented method and the evaluation needed to support the hypothesis. It does not report completed real-data generalization results. Parameters are therefore described as *effective simulation parameters*, conditional on the constitutive law, discretization, geometry, and contact model [8].

II. RELATED WORK

A. Dough mechanics and inverse rheological modeling

Food-engineering studies provide both precedents for inverse simulation and reasons to limit the interpretation of a fitted robot model. Fabbri and Cevoli identify the consistency and flow indices of a power-law semolina-dough model by inverting finite-element extrusion simulations against force-derived pressure measurements [9]. Their comparison with capillary rheometry evaluates a characterization procedure, rather than prediction from partial visual observations. Cocuzza and Yan identify a spring-dashpot model of fondant icing using controlled mechanical tests before evaluating gravity- and tool-induced sheet deformation [10]. These studies show that inverse modeling of food materials predates recent differentiable robotic simulators, and that the selected constitutive equation determines what its parameters mean.

Flour-dough rheology includes nonlinear and history-dependent effects that are not fully represented by a compact elastic-viscous-plastic approximation [2]. Residual deformation after a short recording does not uniquely demonstrate yield plasticity: slow recovery, viscous flow, and interface adhesion can produce similar observations. Adhesive behavior also depends on probe material and test conditions [11]. A principal-stretch limit should therefore be reported as a dimensionless model parameter, not renamed a measured yield stress. For the present study, controlled loading rates, unloading, and recovery observations are important complements to complex two-tool motion.

B. Learned models for elastoplastic manipulation

RoboCraft reconstructs particle representations from RGB-D observations and learns graph-network dynamics for elastoplastic object manipulation [3]. RoboCook extends learned prediction and planning to diverse tools and long-horizon manipulation of real dough [4]. These results establish real food manipulation as an existing robotics application, rather than an unexplored use of deformable simulation. DoughNet predicts geometry and

topological changes from a single RGB-D observation and tool actions [12], demonstrating that limited-view action-conditioned prediction is also studied outside physical parameter identification.

Learned dynamics and compact constitutive identification address related but distinct goals. A predictive network can be useful without assigning a mechanical interpretation to its latent representation. Conversely, explicit coefficients are not automatically more accurate or identifiable. A comparison should therefore match the claim: forward prediction under the same recorded actions, data required for calibration, or consistency with independent mechanical tests. Final target-reaching scores from a planner are not directly comparable to geometric errors from a prescribed-motion replay.

C. Elastoplastic material point identification

DPSI is a close precedent for the proposed inverse problem [5]. It combines fixed-corotated elasticity, von Mises plasticity, MLS-MPM/APIC transfer, and rigid-tool signed-distance contact. A robot executes known trajectories on plasticine; one camera is moved to six viewpoints before and after manipulation, and the endpoint clouds are fused. Thus, one physical camera supplies multi-view endpoints rather than a fixed-view time series. The method fits material parameters and, in a higher-contact-complexity setting, interface friction. It also evaluates unseen, longer tool motions. Known actions, incomplete-volume completion, elastoplastic fitting, and action-transfer evaluation are therefore established contributions. Its reported parameter ambiguities and model-dependent recovery behavior further motivate evaluating prediction separately from parameter uniqueness.

EMPM extends embodied material point modeling to offline demonstrations and online bimanual interaction, explicitly including bread dough and observations from three RGB-D cameras [6]. Its general formulation includes elastic and plastic quantities, while the reported experimental implementation optimizes homogeneous Young’s modulus and Poisson’s ratio. Gradient-based fitting and a forward-simulator CMA-ES option are both described. The online bread-dough experiment evaluates alignment with ongoing parameter correction; this differs from fixing parameters before an independent manipulation. Our proposed comparison concerns the value of time-resolved, single-view evidence and a separately defined rate term, not the novelty of two arms or dough simulation. The principal-stretch projection used here also differs from the von Mises laws in DPSI and EMPM and must be evaluated on its own merits.

D. Visual identification of elastic and dissipative response

Arnavaz et al. fit soft-body simulations directly to depth images from a single camera [7]. With known silicone-specimen geometry and clamp conditions, their dynamic example jointly estimates Young’s modulus, a damping coefficient, and density. They explicitly discuss

dependence on the constitutive model and discretization. Their damping coefficient is not interchangeable with the additive viscosity coefficient used here. This work establishes single-depth-view elastic and dissipative calibration, while leaving different measurement and contact assumptions from an initially incompletely observed dough object.

DiffCloud combines differentiable simulation and rendering with two-camera point clouds and recorded robot motion for cloth and other deformable-object experiments [13]. Its real fitting examples use selected informative frames and a directed simulated-to-observed point-distance loss. The present directed term has the opposite orientation, from observations to visible predicted particles, and is combined with time-resolved depth and coverage penalties. The choice of direction determines which unsupported regions receive a penalty; neither directed distance is equivalent to a symmetric Chamfer loss.

Earlier inverse mechanics studies also inform the evaluation. Wang et al. estimate soft-object models from tracked multi-view motion and include independent loading comparisons [14]. Hahn et al. use marker motion capture and measured clamp motion to identify elastic and power-law viscous response, showing the importance of varied excitation and boundary conditions [15]. These studies support testing multiple motions and numerical resolutions rather than interpreting optimizer convergence as sufficient evidence of material recovery.

E. Visual continuum models and interaction-conditioned prediction

PAC-NeRF couples differentiable material point simulation to radiance-field reconstruction for multi-view system identification [16]. GIC uses Gaussian-informed geometry and temporal visual objectives for continuum-parameter identification and simulation [17]. Both consider multiple constitutive families in synthetic experiments, including dissipative or plastic behavior. Their broader formulation should be distinguished from the particular real experiments that were demonstrated; coverage of a constitutive family in synthetic data does not establish joint rheological identification of manipulated flour dough.

PhysTwin provides a complementary spring-mass approach fitted to multi-view hand-object interactions [18]. It evaluates unseen interactions by reusing the fitted model after registration to the new episode’s first observation. This is a relevant precedent for separating material reuse from initial-state reconstruction. In our proposed evaluation, a held-out episode likewise receives its own first-frame geometry and recorded actions, but its subsequent dough observations must not influence calibration or parameter selection.

F. Numerical foundations and the role of gradients

MLS-MPM supplies the particle-grid discretization [19], while APIC preserves a locally affine velocity representation during transfer [20]. Stomakhin et al.’s snow model

provides a precedent for principal-stretch plastic projection with corotated elasticity [21]; it does not validate that constitutive family for flour dough. ChainQueen, DiffTaichi, and PlasticineLab establish differentiable material point simulation, differentiable programming, and elastoplastic manipulation benchmarks [22]–[24]. Reusing these numerical ideas is a methodological choice, not by itself a research contribution.

Gradients can improve optimization efficiency but do not supply missing experimental information. Different parameter combinations can remain practically indistinguishable under partial observation [25], and a fitted parameter can compensate for systematic model discrepancy [8]. The scientific question is therefore whether the proposed measurements constrain useful predictions across motions, with uncertainty and numerical dependence reported. Endpoint versus time-resolved fitting, restricted versus joint parameter sets, and gradient versus derivative-free search under comparable budgets are appropriate tests of that question.

III. METHOD

A. Problem definition and separation of parameter families

For episode e , the inputs comprise timestamped depth-derived dough points, calibrated camera geometry, prescribed marker poses for two rigid tools, and a recorded object mass. Let s_e^n denote the simulator state and a_e^n the interpolated tool controls at substep n . The discrete forward model is

$$s_e^{n+1} = \Phi_h(s_e^n, a_e^n; \theta, \psi_e), \quad \hat{o}_{e,t} = \mathcal{R}(s_e^{n(t)}; \psi_e), \quad (1)$$

where h is grid spacing, $\mathcal{R}$ is the observation operator, and $n(t)$ associates a measurement with its saved simulation state. The shared fitted parameters are

$$\theta = (E, \nu, \eta, \sigma_{\min}, \sigma_{\max}). \quad (2)$$

They specify elastic stiffness, Poisson’s ratio, the additive viscous coefficient, and lower and upper principal-stretch limits. The fixed episode data ψ_e contain the initial reconstruction, mass, camera calibration, tool geometry and registration, and contact settings. These quantities are not adjusted to reduce the material objective. This separation makes the resulting parameter estimates conditional on explicit geometric and contact assumptions.

B. Table alignment, camera geometry, and tool registration

A robust plane estimate defines the table in the original scene coordinates as $n^\top x + d = 0$, with $\|n\| = 1$ and n oriented toward the object. A rigid transform with rotation $Qn = e_y$ and translation $t_y = d$ yields

$$x_S = Qx + t, \quad (x_S)_y = n^\top x + d. \quad (3)$$

The remaining translation centers the scene within the simulation domain without changing metric scale. Camera extrinsics, observed points, and complete tool trajectories

receive the same transform. The table is then $y = 0$. We take its normal as physical up and set gravity to $[0, -9.81, 0]^\top \text{ m s}^{-2}$; this treats the original tilt as a coordinate-calibration error, rather than a physically inclined table.

Writing T_{AB} for a transform from frame B to frame A , the tool pose is

$$T_{SB}(t) = T_{SM} T_{MK}(t) T_{KB}, \quad (4)$$

where M is the recorded motion frame, K the tool marker, and B the tool's local geometric frame. Mesh units and visual-origin conventions are resolved once when constructing that local geometry. Each constant T_{KB} is estimated separately from color-selected partial tool clouds using visibility-aware robust registration over multiple frames, with fixed camera calibration and recorded marker motion. Multiple initializations and held-out registration frames assess ambiguity. A low point-distance residual alone does not uniquely constrain a partially visible flat tool. The selected transforms are fixed during material fitting.

Tool translations are interpolated linearly and orientations by shortest-path spherical interpolation. The collision-point velocity includes translation and rotation:

$$v_{\text{tool}}(x) = v_c + \omega \times (x - c). \quad (5)$$

Transform composition, metric scale, complete motion bounds, and initial interpolation support are checked before simulation. Raw recordings remain unchanged; derived geometry and configuration files retain content hashes.

C. Initial hidden volume and mass assignment

A single view does not observe the underside or interior. We initialize an approximate solid by voxelizing retained visible points, constructing visible top-height columns, and filling those columns down to the table. Points below a configured table-clearance threshold are removed before reconstruction. A small footprint dilation closes local holes. These operations impose an initial table-contact prior, not an observation of hidden material. They can be inappropriate for overhangs or an initially lifted object.

For occupied-voxel count N_v , voxel width a , particle count N_p , and recorded total mass m , the assignment is

$$\begin{aligned} V &= N_v a^3, & \rho &= \frac{m}{V}, \\ V_p &= \frac{V}{N_p}, & m_p &= \frac{m}{N_p}. \end{aligned} \quad (6)$$

Particles are sampled from this reconstructed volume with a fixed random seed and initialized at rest. *Downward filling is applied only to the initial reconstruction.* During replay, particle positions are determined by dynamics and contact; neither simulated states nor later observations are refilled to the floor. An unexplained simulated floor gap must therefore be diagnosed, not removed by geometric completion.

The current Episode 18 input preparation uses 3 mm voxels, a 3 mm observation clearance, and 24,000 particles. Its 4,216 occupied voxels yield 113.832 mL. Preserving the recorded 250 g mass implies approximately $2,196 \text{ kg m}^{-3}$. This unusually high density requires checking the recorded mass and hidden-volume assumption before fitted coefficients can be interpreted as physical measurements.

D. Constitutive approximation and corrected MLS-MPM transfer

Each particle stores position x_p , velocity v_p , affine velocity matrix C_p , and deformation state F_p . At each substep, the trial deformation and principal-stretch projection are

$$F_p^* = (I + \Delta t C_p^n) F_p^n = U_p \text{diag}(\sigma_p) W_p^\top, \quad (7)$$

$$\bar{F}_p = U_p \text{diag}(\text{clip}(\sigma_p, \sigma_{\min}, \sigma_{\max})) W_p^\top. \quad (8)$$

Here U_p and W_p are the left and right singular-vector matrices. The stored next deformation state is $F_p^{n+1} = \bar{F}_p$. Projection occurs before stress evaluation. The active model has no plastic-volume hardening. This is a Stomakhin-style principal-stretch approximation [21], not a von Mises or J2 return mapping.

With $R_p = \text{polar}(\bar{F}_p)$, $J_p = \det \bar{F}_p$, and

$$\mu = \frac{E}{2(1 + \nu)}, \quad \lambda = \frac{E\nu}{(1 + \nu)(1 - 2\nu)}, \quad (9)$$

the quantity entering the momentum transfer is

$$\begin{aligned} \tau_p &= 2\mu(\bar{F}_p - R_p)\bar{F}_p^\top + \lambda J_p(J_p - 1)I \\ &\quad + \eta(C_p^n + (C_p^n)^\top). \end{aligned} \quad (10)$$

The elastic part is the first Piola stress multiplied by $\bar{F}_p^\top$, a Kirchhoff-type quantity; τ_p is not itself first Piola stress. The final term is the implemented additive rate-dependent approximation. It does not introduce a general relaxation spectrum or establish a complete dough rheology model.

For quadratic B-spline weights w_{ip} and metric offsets $d_{ip} = x_i - x_p^n$, particle-to-grid transfer is

$$A_p = m_p C_p^n - \Delta t V_p \frac{4}{h^2} \tau_p, \quad (11)$$

$$m_i = \sum_p w_{ip} m_p, \quad (12)$$

$$p_i = \sum_p w_{ip} (m_p v_p^n + A_p d_{ip}). \quad (13)$$

Grid velocity is normalized by positive accumulated mass, updated with gravity, and modified by boundary and tool contact. Denoting the resulting velocity by $\tilde{v}_i$, grid-to-particle transfer and advection are

$$v_p^{\text{adv}} = \sum_i w_{ip} \tilde{v}_i, \quad (14)$$

$$C_p^{n+1} = \frac{4}{h^2} \sum_i w_{ip} \tilde{v}_i d_{ip}^\top, \quad (15)$$

$$x_p^{\text{adv}} = x_p^n + \Delta t v_p^{\text{adv}}. \quad (16)$$

Particle contact then produces the final position and velocity. The $4/h^2$ affine factor is required with metric

offsets. Floor-adjacent stencils use floor-based indexing and padded grid support; that numerical padding does not move the physical table.

E. Non-adhesive tool contact and floor response

Rigid-tool contact uses voxelized signed-distance fields (SDFs), with normals from their spatial gradients. For an outward unit normal n , relative velocity $u = v - v_{\text{tool}}$, normal component $b = u^\top n$, and tangent $u_t = u - bn$, the active velocity-level rule is

$$u' = \begin{cases} u, & b \geq 0, \\ \max\left(0, 1 - \frac{\mu_f(-b)}{\|u_t\|}\right) u_t, & b < 0, \|u_t\| > 0, \\ 0, & b < 0, \|u_t\| = 0. \end{cases} \quad (17)$$

The world velocity is $v' = v_{\text{tool}} + u'$. Separating motion is unchanged, and tangential reduction occurs only during inward motion. The fixed $\mu_f = 0.5$ is not fitted with material parameters. Absorption and stickiness are disabled.

This is a discrete Coulomb-like impulse projection using inward velocity as a normal-impulse proxy, not a pressure-based contact solve. Resting friction can be weak when inward speed is small. Grid contact selects a nearest qualifying tool, while post-advection particle contact checks the two tools sequentially and projects penetrating particles outside the SDF by the configured padding plus a small numerical offset. Angular tool velocity is evaluated at the projected contact location. The floor uses a different rule: position is constrained to $y \geq 0$, inward normal velocity is removed, and tangential retention is fixed at 0.4. That retention factor is not a Coulomb coefficient.

F. Differentiable partial-view observation model

Particle positions are transformed into calibrated optical coordinates $q_p = (x_p^c, y_p^c, z_p)$ and projected to image coordinates (u_p, v_p) . For pixel k , a compact footprint with radius r is

$$f_{pk} = (1 - \delta u_{pk}^2)^2 (1 - \delta v_{pk}^2)^2, \quad |\delta u_{pk}|, |\delta v_{pk}| < 1, \quad (18)$$

with zero weight outside its support. The normalized offsets divide projected pixel-center differences by r . Depth-dependent weights are

$$a_{pk} = f_{pk} \exp\left[-\frac{z_p - z_{\text{front},k}}{T}\right], \quad (19)$$

where $z_{\text{front},k}$ stabilizes the exponential numerically and cancels from the normalized ratios. Predicted depth and coverage are

$$\begin{aligned} \hat{D}_k &= \frac{\sum_p a_{pk} z_p}{\sum_p a_{pk}}, \\ c_k &= 1 - \exp\left[-\gamma \frac{\sum_p a_{pk} f_{pk}}{\sum_p a_{pk}}\right]. \end{aligned} \quad (20)$$

Pixels without support have zero coverage and are marked invalid for depth. Coverage is based on a normalized mean footprint, not accumulated volumetric opacity. The

current settings are $r = 2$ pixels, $T = 5$ mm, and $\gamma = 8$, using 160×120 images. Projection clipping and footprint membership remain piecewise operations. The dough renderer does not explicitly model occlusion by tools, so tool-obscured measurements remain a limitation.

G. Depth-change, coverage, and visible-point objective

Let M_t^o and M_t^s denote valid observed and predicted depth masks. The depth-change comparison uses

$$\begin{aligned} S_t &= M_t^o \cap M_0^o \cap M_0^s, \\ r_{tk} &= \frac{(\hat{D}_{tk} - \hat{D}_{0k}) - (D_{tk} - D_{0k})}{s_d}. \end{aligned} \quad (21)$$

Crucially, S_t does not require current predicted support. On pixels with current support, define

$$\ell_{tk} = c_{tk} \rho_H(r_{tk}) + (1 - c_{tk}) \rho_H(r_{\text{miss}}), \quad (22)$$

where $\rho_H(r) = r^2/2$ for $|r| \leq 1$ and $|r| - 1/2$ otherwise. On pixels without current support, define $\ell_{tk} = \rho_H(r_{\text{miss}})$ directly. Thus, losing the predicted object does not remove its observed pixels from the depth penalty. With $s_d = 10$ mm and $r_{\text{miss}} = 4$, the missing-prediction penalty is 3.5. The first two components are

$$L_{d,t} = \frac{1}{|S_t|} \sum_{k \in S_t} \ell_{tk}, \quad L_{c,t} = \frac{1}{|M_t^o|} \sum_{k \in M_t^o} (1 - c_{tk}). \quad (23)$$

These terms supervise observed object regions without declaring unknown pixels to be empty space.

A third term compares observed optical points y_j with predicted particles classified as visible. A particle is visible when its depth is within $2T$ of the front depth at its projected-center pixel. Defining $p^*(j)$ as its nearest visible predicted neighbor and using $s_p = 10$ mm,

$$L_{p,t} = \frac{1}{M_t} \sum_{j=1}^{M_t} \rho_H\left(\frac{\|q_{p^*(j)} - y_j\|}{s_p}\right). \quad (24)$$

At most 1,024 observed points are selected by deterministic index subsampling. This is a directed observation-to-visible-prediction loss, not symmetric Chamfer distance. Visibility and nearest-neighbor choices are fixed during the local backward calculation, while particle coordinates are differentiated.

Configured component weights (1, 0.5, 0.5) are normalized to give

$$L_t = 0.5L_{d,t} + 0.25L_{c,t} + 0.25L_{p,t}. \quad (25)$$

For K_e scored observations in episode e , shared-material identification minimizes

$$\begin{aligned} L(\theta) &= \sum_{e \in \mathcal{E}_{\text{train}}} \alpha_e L_e(\theta), \\ L_e &= \frac{1}{K_e} \sum_t L_{e,t}, \quad \alpha_e = \frac{w_e}{\sum_j w_j}. \end{aligned} \quad (26)$$

This averages each episode once over time and then applies fixed positive episode weights. Longer recordings do not automatically dominate an equal-episode-weight objective.

Empty or insufficient required observation support rejects an evaluation instead of silently dropping a frame or episode.

H. Checkpointed differentiation and bounded optimization

Reverse-mode differentiation propagates the observation gradients through the complete discrete rollout. Full particle states are checkpointed at segment boundaries and recomputed during the backward pass. Adjoint values for position, velocity, affine velocity, deformation, and any enabled plastic-volume state are carried across boundaries; observation gradients are injected once at their corresponding saved states. Segment length controls recomputation storage, not gradient truncation.

Custom vector-Jacobian products differentiate the composite polar and principal-stretch projection maps, including repeated singular values away from projection thresholds, rather than differentiating the internals of an iterative SVD. Grid normalization has a dedicated adjoint to avoid unstable intermediate divisions for very small positive masses. Contact switches, visibility selection, nearest neighbors, and clamp thresholds make the overall objective piecewise differentiable. Chosen threshold sensitivities are numerical conventions at points without an ordinary derivative.

Recomputed states, losses, and contact diagnostics are checked against the forward pass. Strict consistency checking is the default. An explicit optional override permits finite mismatches, as can occur with parallel floating-point accumulation, and labels the resulting gradients approximate. Invalid states, nonfinite quantities, and invalid observation support remain errors. This distinction is necessary when interpreting GPU optimization results.

Projected Adam operates in scaled coordinates

$$\begin{aligned} z_E &= \log \frac{E}{E_{\text{ref}}}, & z_\nu &= \frac{\nu}{0.1}, \\ z_\eta &= \frac{\eta}{100 \text{ Pa s}}, \\ z_- &= \frac{1 - \sigma_{\min}}{0.1}, & z_+ &= \frac{\sigma_{\max} - 1}{0.1}, \end{aligned} \quad (27)$$

where E_{ref} is a fixed coordinate reference. Bounds and initial values are given in Table I. Each proposed update is projected into the admissible interval and evaluated by a new rollout. Backtracking reduces the step until a valid candidate satisfies sufficient decrease. Rejected proposals leave parameters and optimizer moments unchanged. The defaults use learning rate 0.05, Adam moments (0.9, 0.999), and at most eight backtracks with factor 0.5.

For multiple episodes, a host-side optimizer evaluates one episode subprocess at a time and aggregates physical-parameter gradients using the weights in (26). Every candidate, including backtracking candidates, evaluates every training episode. A failed episode invalidates the candidate; its weight is not redistributed. Shared fitting assumes the recordings plausibly share material condition,

TABLE I
CURRENT EXPERIMENTAL CONFIGURATION, NOT FITTED RESULTS.

Quantity	Initial / fixed value	Fitted bounds
E	130.579 kPa	10–300 kPa
ν	0.30	0.15–0.45
η	0 Pa s	0–100 Pa s
$\sigma_{\min}$	0.90	0.70–0.999
$\sigma_{\max}$	1.10	1.001–1.30
Grid spacing h	1/48 m	Fixed
Time step	0.2 ms	Fixed
Tool padding	$h/8$	Fixed
Tool friction μ_f	0.5	Fixed
Floor retention	0.4	Fixed

rather than treating different batches or preparations as automatically equivalent.

I. Evaluation scope and interpretation

The evaluation separates numerical derivative checks, agreement on fitted recordings, and prediction under frozen parameters. Small synthetic finite-difference tests assess the implemented adjoint, but do not validate a physical dough model. A real-data study should compare initialization against calibration, endpoint against time-resolved loss, restricted against joint parameter sets, and single- against multi-episode fitting. Multiple initializations and numerical-resolution checks are needed to determine whether different low-loss coefficients imply similar predictions. A gradient-versus-derivative-free comparison should account for forward and backward computational cost, not only iteration count.

For an independent episode test, shared material parameters and global camera, table, and tool calibrations must be fixed using separate training or calibration data. The new episode is initialized only from its first dough observation; recorded tool trajectories remain legitimate prescribed inputs. Future dough observations are used only for evaluation. Geometric errors, coverage, contact behavior, and recovery should be reported alongside the fitting objective. Later frames from a fitted episode test temporal continuation, not independent action transfer.

The prepared Episode 18 experiment fits frames 1–60 and scores frames 61–97 from a rollout beginning at frame 0. These later frames are held out from *material optimization only*: geometric preprocessing and registration use data across Episode 18. They do not constitute an independent end-to-end test. Input validation has passed for the registered-tool configuration, but its real CUDA gradients, material fit, and predictive accuracy remain unestablished. The density discrepancy and the floor gap observed in an earlier replay with different tool transforms also require investigation. These limitations define what must be checked before claiming that fitted coefficients predict unseen dough behavior or represent independently measured material properties.

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